



Student guide to the Bachelor's project

Purpose of the Bachelor's project

The Bachelor's project is the final phase of the Bachelor's degree programme. The project and the thesis that relates to it are partly designed to assess your academic/scientific knowledge and skills. The project also has another important learning outcome, namely being able to carry out a substantial research project independently. This is something else that you need to learn.

During the Bachelor's project you will learn how to conduct research independently and report on your findings. You will learn how to familiarize yourself with new theories and techniques, how to build and implement a model and/or how to conduct experiments and analyse the data they generate. As a 'finishing touch', you will also learn to write an academic report on the whole process in the form of a thesis and oral presentation. Ultimately, you will be assessed on the product you submit. Your thesis is the most important thing, but you will also be assessed in terms of the process and your approach and work method.

Different from other course units

The subject of the Bachelor's project will partly relate to the knowledge and skills you acquire during your degree programme, but you will also have to acquire a lot of new knowledge and skills during the project. You will have to do this independently and in consultation with the supervisor. The knowledge will not be 'provided' for you; you will have to find it for yourself. The literature on your subject area is an important source of knowledge. Obviously, your supervisor is an important source too. The learning outcome of the project is therefore to learn to do research independently. You will learn to gather knowledge, set up a functional model, conduct experiments, analyse data and, finally, write an academic report on the research. The subject of your research will be very specific and you will be an expert on the subject when you have completed the project.

Learning process

Because this is the first time that you will be doing research, the standard will not be set too high. You may expect the supervisor to set out much of the project for you and provide the necessary basic knowledge. Using a fairly specific project setup and a number of academic articles as a foundation, you will be able to carry out the further work on your project. Remember that this is a learning process. You are not expected to get everything right immediately. You are allowed to make mistakes! So do not hesitate to contact your supervisor if you have questions. 'Independently' does not mean that you are not allowed to ask for help. In fact, recognizing when you need assistance is part of working independently.

Experiences

This guide is based on a guide that an AI staff member wrote for his Bachelor's project students. He noticed that there were some things that his students did not know or understand, and wrote this guide to help them. It is therefore based on many experiences with earlier Bachelor's projects. We hope that it will help you too!

In this guide

This document is about working with the supervisor of your Bachelor's project. It does not contain information about seminars and the related assignments. You will be informed about these during the seminars. The document does contain a number of points that you need to bear in mind when you are doing your research.

First, a list of the **steps** in the project is provided. This will give you an idea of what the Bachelor's project involves. This is followed by a list of **mutual expectations**, i.e. what is expected of you and what you may expect of your supervisor.

There is also a list of **interim products** that you must submit to your supervisor. This is so that your research proceeds in a structured way. The interim products include the sections of your thesis, which you will write throughout the project. This ensures that you do not have to do all the writing at the end. Experience has shown that the writing always takes longer than expected.

The next section in this guide is a description of the requirements for the **thesis content**: what should it include, and where? This is followed by a list of **assessment criteria**. It is good to bear this list in mind throughout the project.

The section after this is a series of **tips** that are based on several years of experience with Bachelor's projects. The document ends with an **overview of documents** that contain more information relating to the Bachelor's project.

What does the project involve?

Drawing up a schedule and deadlines

What are you going to do, and when? When are the deadlines? A list of interim and final products can be found below. You must include the deadlines in your planning.

Research question

Defining the research question and the objective of the project.

Literature study

Taking the recommended literature as a starting point, you must find the literature you need for the research. This may be literature in which certain methods are discussed, or studies on the same subject.

Setting up the model / algorithm / neural network / ...

On paper, produce an outline of the model / algorithm / neural network / (...) you will implement. What functional elements does it consist of, and how do they relate to each other?

Implementing the model / algorithm / neural network / ...

Designing experiments

How are you going to test the research question? Which data should be collected? Think about graphs you may need to make later. How will you gather the data?

Conducting experiments

Analysing the data

What are the results of the experiment(s)? Are trends/effects visible? How can their significance be tested?

Writing the thesis

This is a continuous process. The first chapter (introduction) is written after you have defined your research question and studied the literature. The second part is written when you have implemented the model and designed the experiment(s). The rest of the thesis is written when you have analysed the results.

What is expected of you?

- Independent study of the literature and the techniques you will use.
- 'Independent' also means recognizing when you need help from the supervisor. If you encounter problems, do not struggle with them for too long. Contact the supervisor.
- Acquire the skills needed to implement the model / algorithm / neural network / (...), conduct the experiments and do the analyses.
- Contribute your own input and creativity to the project.
- Good reporting and discussions with the supervisor. Aim to meet with the supervisor once a week or once every two weeks.
- Submit texts in time for scheduled meetings so that the work can be read, then discussed at the meeting.
- Make notes during the meetings and send them to the supervisor afterwards. This is to make sure that students' and supervisors' ideas correspond.

What can you expect of your supervisor?

- Help with all types of problems (literature, implementation, experiments, analysis, textual).
- Regular meetings (usually every week or every two weeks).
- Reading, marking and commenting on submitted texts.
- Final assessment of the project (thesis and process).

Which products do you have to submit, and when?

[This is an **example**. Each supervisor may require different products.]

When?	What?
After first meeting	– Scheduling, including deadlines for the products – Research question, half an A4, including brief background and reasons for the project
After literature study	– Literature survey, including brief summaries – Text of first chapter of thesis (introduction)
Before implementation	– Structure of the model. Identify the functional elements – Proposal for experiments, data gathering
After implementation	– Text of second chapter, methods – Demonstration of functioning model
Before experiment	– Experimental setup and analysis proposal
After experiment	– Text of third chapter, results and analysis
Final phase	– Draft version of oral presentation
Final phase	– Give oral presentation
Final phase	– Draft version of complete thesis. Review of all chapters, add discussion and summary
Final phase	– Thesis

Thesis content

(see also Report Template)

Summary

The summary is a brief description of the subject of the research, how the research was carried out, and what the main results are. The summary is written last.

Introduction

Description of the objective of the project/thesis. Indicate how the research is relevant. Embedding of the subject in the literature (backgrounds). Discussion of similar research. Indicate what was researched in the project and how it relates to other research.

Methods

How are you going to answer the research question? This is often a matter of describing the model / algorithm / neural network / (...) used/implemented. The experimental setup is also discussed. Explain clearly why the experiments are important for answering the research question.

Results

Factual reporting on experiment results. Generally speaking, graphs are very important here. The results must also be thoroughly analysed. Statistical tests are often used for this purpose.

Discussion

In this phase, the significance of the results is discussed. When analysing the results, you also need to look for explanations. Why are the results as they are? This increases understanding of the methods used, with reference to the research question. How can the question be answered? How do the model used and the results relate to other research? At this stage, deficiencies or problems can be discussed. Finally, what is the conclusion of the research? Suggestions for future research can be made.

Assessment

You will be assessed on the quality of your report, as you would be in the world of science. It is important to demonstrate that you have explored the subject thoroughly. Your research, oral presentation and work process will also be assessed, but these aspects are less important than the report. In principle you will be given a mark of 7 if you have completed the assignment to a reasonable standard. You can obtain a higher mark by doing more than is required (e.g. showing a lot of initiative) or by writing a good report.

There is an **assessment form** for the Bachelor's project. The criteria on the form include individual input, approach, independence, the oral presentation, the thesis and the research itself.

More information/documents

Below you will find a list of sources with more information about the Bachelor's project, and documents that you may find useful during the project. All this information (and more) is available on the Nestor page for the Bachelor's project.

Regulations for the AI Bachelor's project

The requirements and work method for the Bachelor's project.

Report template for Bachelor's project

The structure of the report on the Bachelor's project: what information it should contain, and how the information should be presented. You must use this template to write your report.

Contract relating to the AI Bachelor's project

This contract sets out the agreements that the student and supervisor have made regarding the Bachelor's project. These are agreements on matters such as the frequency of supervisory meetings. In the Appendix you can indicate what agreements have been made regarding project planning and deadlines.

Assessment form

This is the form that your supervisor will use to assess your work. It is useful to look at this to see how your work will be assessed.

Tips

- When you do assignments for the Bachelor's project, let your supervisor see them too. Your supervisor will not always know what assignments you have to do.
- Ask your supervisor questions. Your supervisor knows a lot about your project, and the idea is that you learn a great deal from him/her.
- Be proactive; take the initiative and put your own ideas forward. Supervisors appreciate this very much.
- However, take care not to be too obstinate. Your supervisor knows more about research and science than you do and is better able to judge what is feasible and what is not.
- Supervisors are often very busy and may forget things. Adopt a constructive approach, and take the initiative if you think your supervisor has forgotten something (e.g. to make an appointment).
- Work at the University, not at home. The environment at the University is more stimulating and more conducive to working effectively, it is easier to call in on your supervisor, and there is greater involvement.
- The supervisor invests time in your project and it is in his/her interest that you achieve good results. That is why it is important for you to be honest about your progress, even if things are not going very well.
- If you have problems with your supervisor, discuss them with the coordinator/lecturer of the Bachelor's project.